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SUMMARY

Software Engineer with a strong systems programming foundation (C/C++, Go) and hands-on full-stack experience across React, Node.js, FastAPI, and Flutter. Currently building a production-grade multi-tenant ERP system with a 400+ table MS SQL Server schema and an orchestrator-driven Node.js backend, and have published open-source tooling extracted from it. Experienced in AI/ML integration, REST API design, and cross-platform mobile development. Seeking to bring both depth and versatility to a high-performance engineering team.

EXPERIENCE

Founder & Lead Engineer — Multi-Tenant SaaS ERP (*Self-initiated SaaS Product*)

Jan. 2026 – Present

- Architecting a **multi-tenant SaaS ERP** with domain-driven schema organization and row-level tenant isolation on a **400+ table MS SQL Server** schema covering core business modules, with domain-specific HMS tables planned post-core
- Engineering a **Node.js/Express.js** backend using an **orchestrator pattern** with use-cases, steps, engines, and pipelines for modular, scalable business logic
- Designing the system as a **commercially distributable product** with multi-tenancy and reseller support built into the core architecture

PROJECTS

penstock — Open-Source npm Library

TypeScript, Vitest, tsup, CI/CD

- Published a **zero-dependency TypeScript library** on npm for composable backend workflows using a **use-cases → pipelines → steps → engines** architecture, extracted and generalized from my production ERP's orchestration layer
- Implemented **saga-pattern rollback/compensation** with automatic reverse-order undo on failure, returning structured results with full step-level visibility; built with **full TypeScript generics** and type-level testing
- Engineered **dual ESM/CJS packaging**, **115 tests** at 95%+ coverage, CI matrix testing (Node 20/22/24), and supply-chain hardening (**OIDC trusted publishing**, prototype-pollution resistance, zero telemetry)

Interpreter in Go

Go, Compiler Design, AST, TDD

- Implemented a complete interpreter for the **Monkey language** in Go, covering lexical analysis, parsing, AST construction, and tree-walking evaluation
- Applied compiler design principles including **Pratt parsing**, scope handling, closures, and first-class functions, using **test-driven development** throughout

MacroMate [FYP]

Flutter, FastAPI, Python, Gemini API, Pretrained ML Model

- Developed an **AI-powered cross-platform mobile app** (Flutter) for calorie, macronutrient, and workout tracking with personalized recommendations
- Built **REST APIs with FastAPI** for authentication and data management; integrated **Gemini API** for food image recognition and a **pretrained model** for meal plan generation

OCR Scanner with AI Summarization

Python, OpenCV, Gemini API

- Built a **Python-based OCR pipeline** achieving **95%+ text extraction accuracy** on multi-page PDFs with sub-3-second processing, using **OpenCV** and **Gemini API** for AI-powered summarization

SKILLS

Languages : C/C++, Go, Python, JavaScript, TypeScript, SQL, Dart, Java

Backend : Node.js, Express.js, FastAPI, REST API Design, Orchestrator Architecture

Frontend : React, Redux Toolkit, TailwindCSS, Flutter

Databases : MongoDB, MySQL, Microsoft SQL Server

Tools : Git, Docker, Postman, Vitest, CI/CD (GitHub Actions)

EDUCATION

University of Central Punjab

Bachelor of Software Engineering

Mar. 2022 – Mar. 2026

CGPA: 3.41

CERTIFICATIONS

Supervised Machine Learning

DeepLearning.AI – Coursera